DESIGN COMPUTED · FREE GUIDE

Topology optimisation: let the forces draw the structure

The free guide to the reel “These steel trees hold up a 250 m roof”: how a computer finds where material is needed, where it is used in real buildings, and how to try it yourself for free.

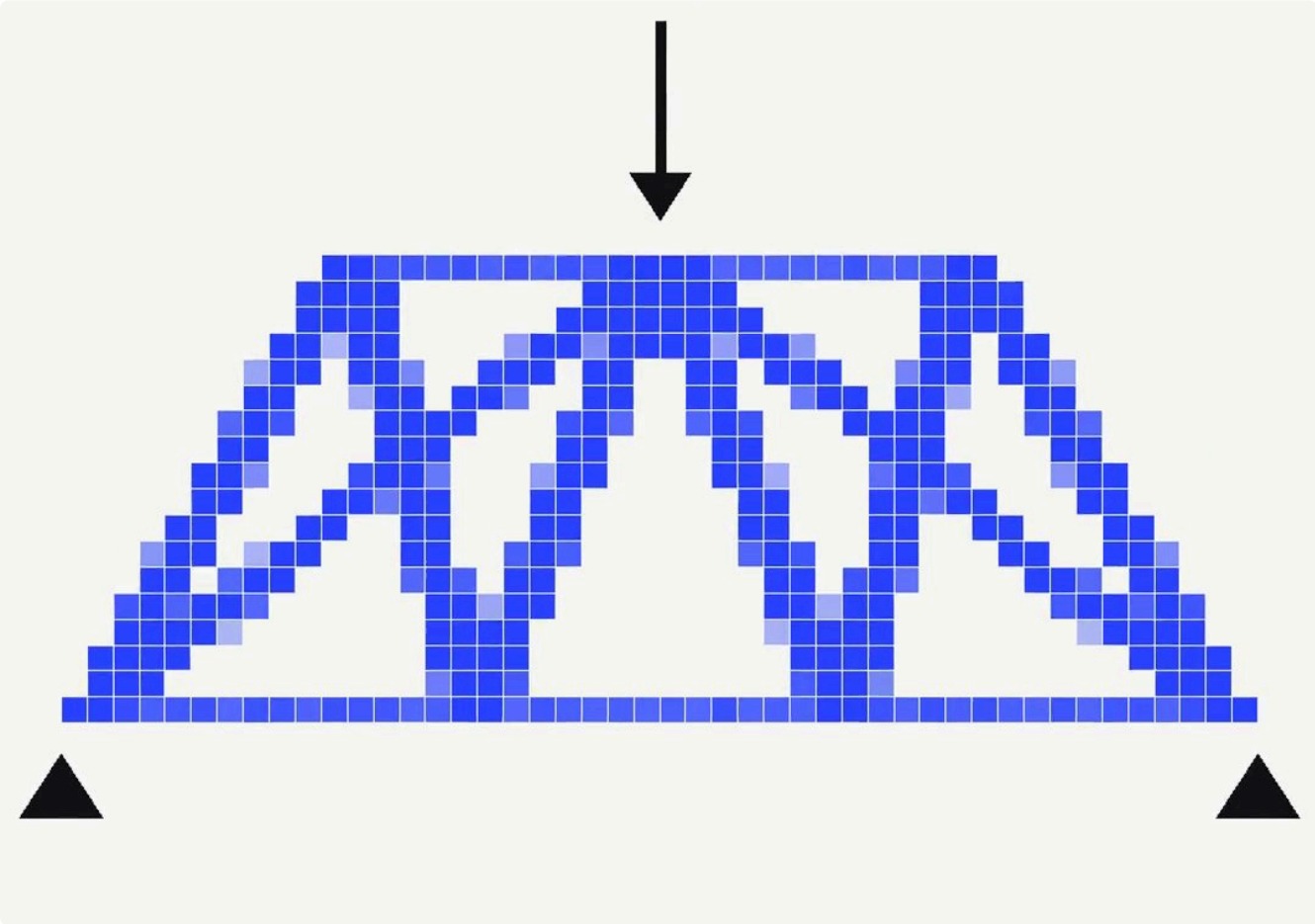


How it works

Topology optimisation answers one question: if I can only use a limited amount of material, where should it go? You give the computer a design space (a block), the supports, the loads, and a material budget, for example 40 % of the block.

The computer cuts the block into thousands of small cells and calculates how hard each cell works under the loads. Cells that carry little force are made weaker or removed; cells on the main force paths are kept. It repeats this, step by step, until the shape stops changing. What is left follows the flow of forces, which is why the results often look like trusses, branches or bones.

Two families of methods are common. SIMP (Bendsøe, 1989) gives each cell a density between empty and solid and pushes it towards 0 or 1. ESO and BESO (Xie and Steven, from 1993) remove the least useful cells and, in the bidirectional version, add material back where it is needed.

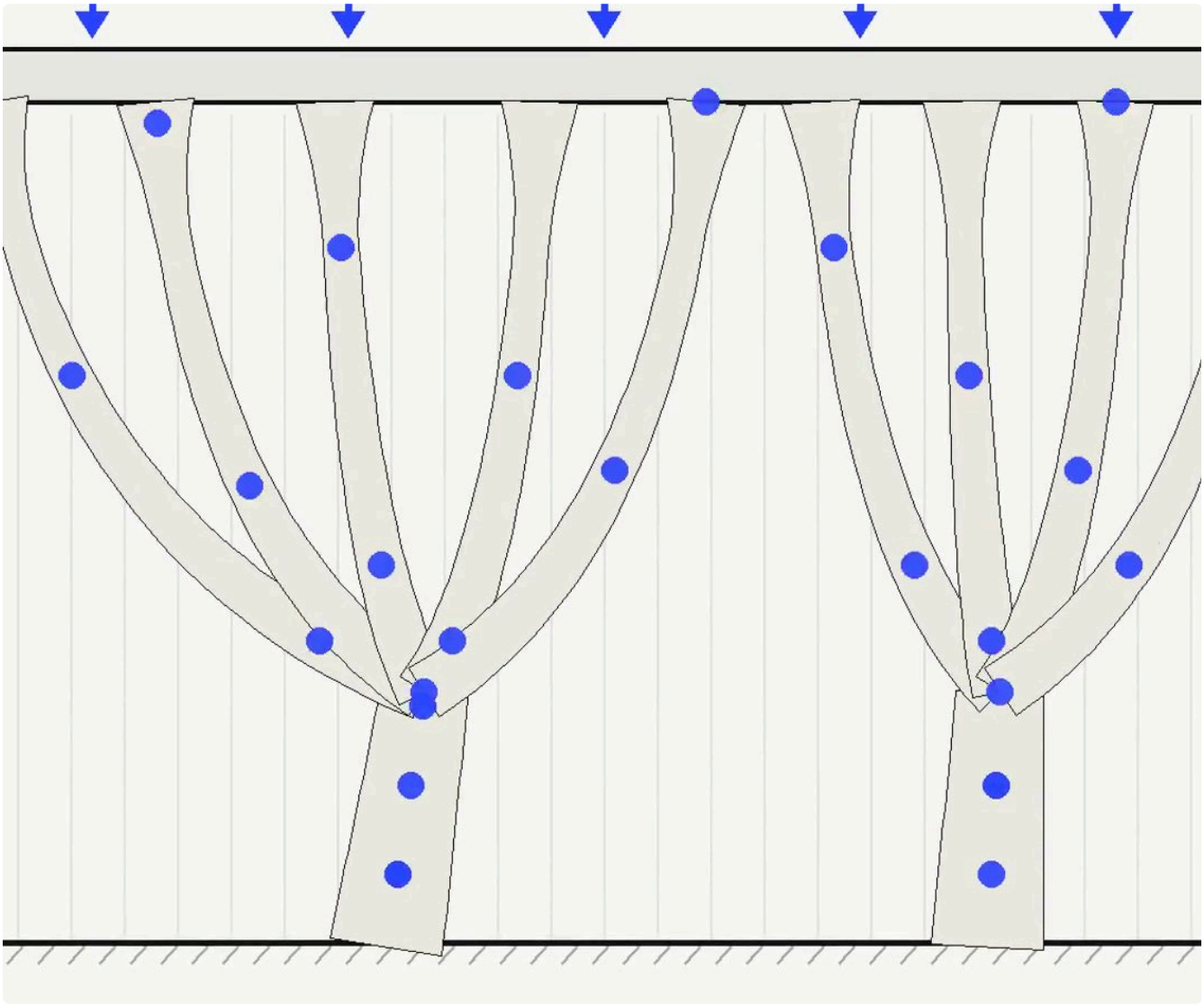


A block with two supports and one load: the computer removes what does not carry force, and a truss appears.

Why it looks like bone

In 1892 the surgeon Julius Wolff described what is now called Wolff's law: bone rebuilds itself along the loads it carries. The spongy bone inside a femur is a lattice of thin struts aligned with the stresses, light where little force passes and dense where a lot does. Topology optimisation follows the same logic with a calculation instead of a living cell.

In Doha, the 250 m façade of the Qatar National Convention Centre (Arata Isozaki, opened 2011) is carried by two steel trees inspired by the sidra tree. Its structural shape was developed with the engineer Mutsuro Sasaki, known for computer methods that grow structures by adding material where stress is high and removing it where it is low.



The Doha trees: every branch follows a load path from the roof to the ground.

Built examples

Qatar National Convention Centre, Doha (2011)	Arata Isozaki with Mutsuro Sasaki: the two steel “sidra” trees of the main façade.
3D-printed steel node, Arup (2015)	Arup’s design team optimised a structural node for a tensegrity structure and printed it in steel: lighter and smaller than the classic welded version.
Bone Chair, Joris Laarman (2006)	A chair shaped with software derived from how bones grow, originally developed for the car industry.
Airbus “bionic partition” (2015)	An aircraft cabin partition designed by generative design with Autodesk, much lighter than the earlier part.
Kakamigahara crematorium, Japan (2006)	Toyo Ito with Mutsuro Sasaki: a concrete roof whose free form was found by optimisation (shape optimisation, a close cousin of topology optimisation).

Try it yourself (free)

1. Open the free TopOpt apps of the Technical University of Denmark (topopt.mek.dtu.dk, “Apps and software”). No installation needed for the 2D web version.
2. Draw a long rectangle: this is your design space, a beam.
3. Add two supports at the bottom corners and one load pushing down in the middle of the top edge.
4. Set the material budget to about 40 % and start. Watch the block turn into a truss in a few seconds.
5. Move the load to one side, or add a second load, and run it again: see how the structure reorganises.
6. Lower the budget to 20 %: the bars get thinner and the shape gets more “bone-like”.
7. In Rhino/Grasshopper, the free plug-in Millipede does the same in 2D and 3D, so you can turn the result into a model.
8. Redraw the result as clean lines: an optimised shape is a starting point for design, not a finished building.

The rule to remember

Put material only where the force goes. Start from the loads and the supports, let the forces draw the first sketch, then design on top of it.

Mistake 1: optimising a bad problem	Wrong supports or loads give a perfect answer to the wrong question. Check them first.
Mistake 2: building the raw result	Optimised shapes need smoothing, buckling checks and joints that can really be made.
Mistake 3: one load case only	Wind, people and self-weight change. Optimise for several load cases at once.

Go further

Topology Optimization: Theory, Methods and Applications — M. P. Bendsøe and O. Sigmund (Springer, 2003)
https://link.springer.com/book/10.1007/978-3-662-05086-6

Efficient topology optimization in MATLAB using 88 lines of code — Andreassen et al. (2011), free code
https://www.topopt.mek.dtu.dk/apps-and-software

Evolutionary Structural Optimization — Y. M. Xie and G. P. Steven (Springer, 1997)
https://link.springer.com/book/10.1007/978-1-4471-0985-3

Millipede for Grasshopper (free) — Panagiotis Michalatos
https://www.food4rhino.com/en/app/millipede

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